

Computational perspectives on the Classical Decision Problem

Lecture 10: A Glimpse Beyond ...

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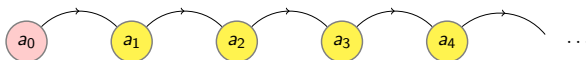
ESSLLI 2026

- We have seen that **transitivity** is not expressible in $\mathcal{FO}^2$.

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- For suppose t is interpreted as a transitive relation. Then

$$\forall x \exists y \, t(x, y) \wedge \forall x \neg t(x, x)$$

is satisfiable, but only in infinite structures:

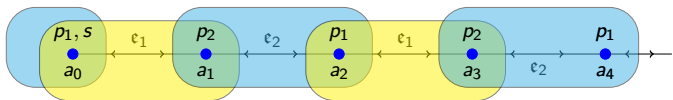


- **Equivalence** is likewise not expressible in $\mathcal{FO}^2$.

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- For suppose ϵ_1 and ϵ_2 are equivalence relations. Then

$$\begin{aligned} & \exists x (p_1(x) \wedge s(x)) \wedge \forall x (p_1(x) \rightarrow \neg p_2(x)) \\ & \bigwedge_{i=1}^2 \forall x (p_i(x) \rightarrow \exists y (\epsilon_i(x, y) \wedge p_{3-i}(y) \wedge \neg s(y))) \\ & \bigwedge_{i=1}^2 \forall x \forall y (\epsilon_i(x, y) \rightarrow (p_i(x) \wedge p_i(y) \rightarrow x = y)) \end{aligned}$$

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- We know from earlier remarks that $\mathcal{FO}^21T$ and $\mathcal{FO}^22E$ lack the FMP.

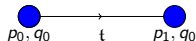
- Having access to transitive relations allows us to ‘snap’ squares into place.
- Consider the following formulas:

$$\exists x(p_0(x) \wedge q_0(x))$$

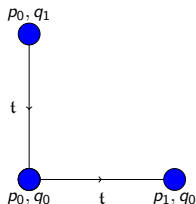

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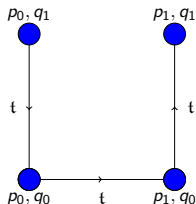


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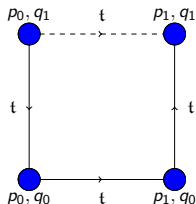
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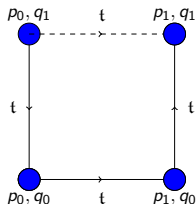
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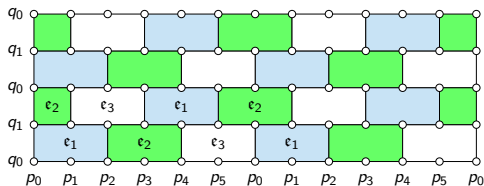
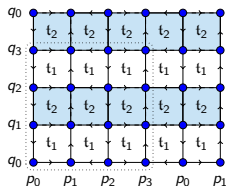
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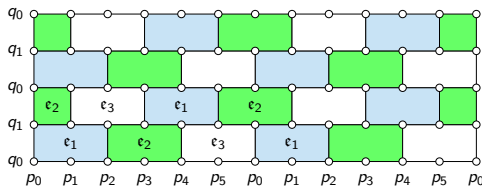
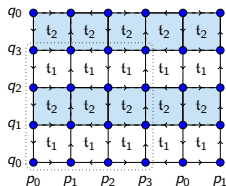
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- A similar situation holds for equivalence relations.

- With two transitive relations or three equivalence relations, we can write satisfiable formulas whose models embed infinite grids:

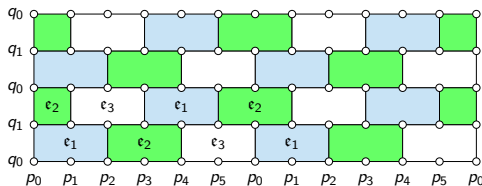
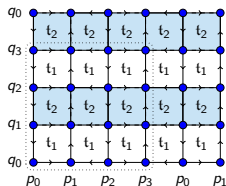


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- This allows us to reduce unbounded tiling problems to $\text{Sat}(\mathcal{FO}^2\text{2T})$ and $\text{Sat}(\mathcal{FO}^2\text{3E})$.

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- This allows us to reduce unbounded tiling problems to $\text{Sat}(\mathcal{FO}^2\text{2T})$ and $\text{Sat}(\mathcal{FO}^2\text{3E})$.
- Hence $\text{Sat}(\mathcal{FO}^2\text{2T})$ and $\text{Sat}(\mathcal{FO}^2\text{3E})$ are undecidable. Similarly for finite satisfiability.

- However, if k is small, (finite) satisfiability can be decidable.

$\mathcal{L}$	$\text{FinSat}(\mathcal{L})$	$\text{Sat}(\mathcal{L})$
$\mathcal{FO}^21\text{T}$		—
$\mathcal{FO}^22\text{T}$	undec.	undec.
$\mathcal{FO}^21\text{E}$		
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$\mathcal{FO}^23\text{E}$	undec.	undec.

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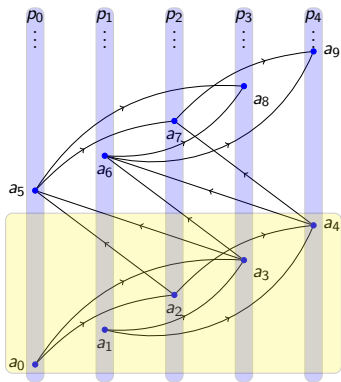
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- The decidability of problem $\text{Sat}(\mathcal{FO}^2\text{IT})$ is open. Infinite models can be forced even when all witness requirements are 'free'.



$p_0, \dots, p_4$ partition the domain

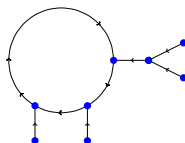
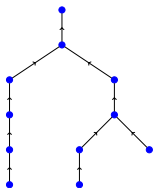
$$\bigwedge_{k=0}^4 \forall x (p_k(x) \rightarrow \exists y (p_{[k+1]}(y) \wedge \neg t(x, y) \wedge \neg t(y, x)))$$

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- We can ‘almost’ say this in $\mathcal{C}^2$. Reading $\mathfrak{d}(x, y)$ as “ x is a daughter of y ” and $\mathfrak{r}(x)$ as “ x is a root”, we can write



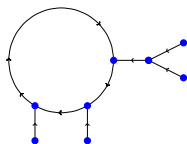
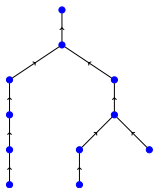
$$\forall x \forall y (\mathfrak{d}(x, y) \rightarrow \neg \mathfrak{d}(y, x))$$

$$\forall x \exists_{[\leq 1]} y \mathfrak{d}(x, y)$$

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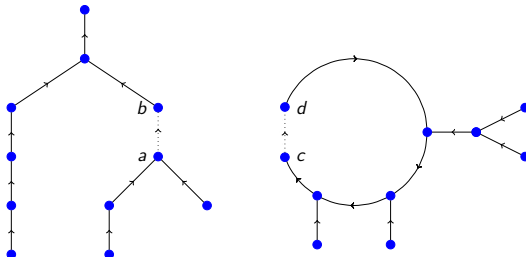
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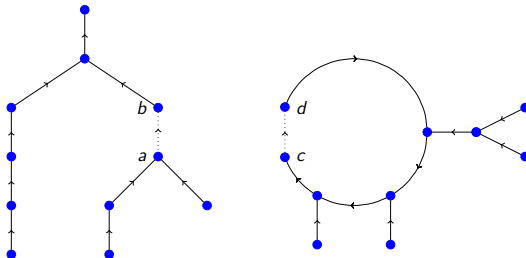
- The problem is that such models can have ‘rootless’ components.

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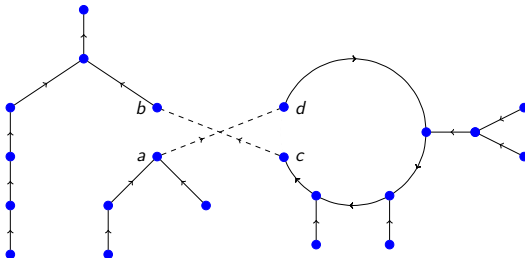


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- Then a and c can switch mothers, this splicing the rootless component into the rooted component.

- Thus, given a $\mathcal{C}^2$ formula φ , we can compute a $\mathcal{C}^2$ formula ψ_φ such that φ is satisfiable in a tree-like model if and only if $\varphi \wedge \psi_\varphi$ is finitely satisfiable.

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- C. and W.'s proof works also with two (independent) tree predicates $\mathfrak{d}_1$ and $\mathfrak{d}_2$.
- However, decidability of $\text{Sat}(\mathcal{C}^2 kD)$ for $k \geq 3$ is open.

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- There are several interesting decidability results here. The locus classicus is [S. Benaim, M. Benedikt, W. Charatonik, E. Kieronski, R. Lenhardt, F. Mazowiecki and J. Worrell ICALP 2013](#).

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$\mathcal{FL}2\mathcal{T}$	-	-
$\mathcal{FL}2\mathcal{T}_=$	undecidable	undecidable
$\mathcal{FL}3\mathcal{T}$	undecidable	undecidable

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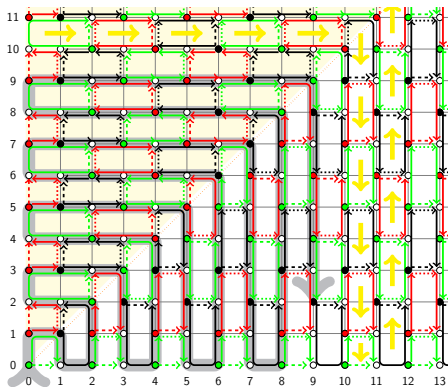
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- The undecidability proofs for $\mathcal{FL}2T_=$ are routine, but not for $\mathcal{FL}3T$.

- Three transitive relations suffice to encode a grid, even without equality or counting:

$$\forall x_1 (c(x_1) \rightarrow \exists x_2 (T_{\text{green}}(x_1, x_2) \wedge d(x_2)))$$

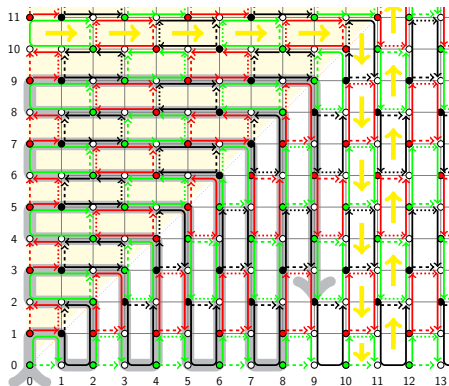
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L. Tendera

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- or even

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- A natural question is whether $\mathcal{C}^2$ can be extended with modulo counting.

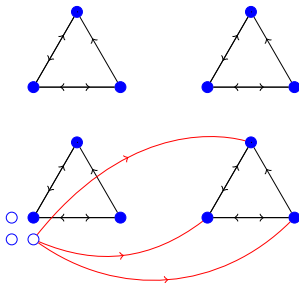
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- The following was shown by M. Benedikt, Egon Kostylev and T. Tan (2025):

Theorem

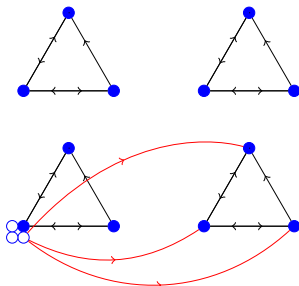
The satisfiability and finite satisfiability problems for $\mathcal{FO}_{Pres}$ are in 2-NEXPTIME.

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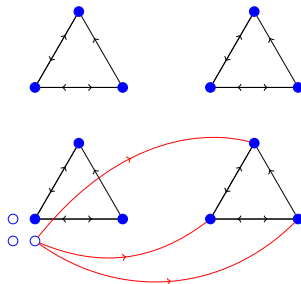


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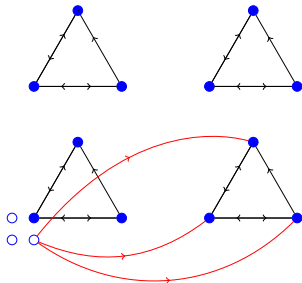


- Then merge the ghost elements with suitable real elements.

- We can then assert requirements on the ‘real’ and ‘ghost’ elements separately, using ordinary $\mathcal{C}^2$ -formulas.

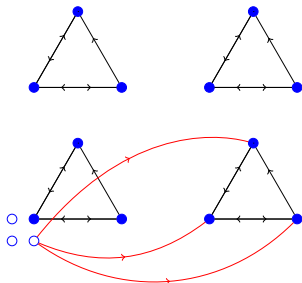


- We can then assert requirements on the 'real' and 'ghost' elements separately, using ordinary $\mathcal{C}^2$ -formulas.



- We have to take elaborate precautions to ensure that ghost elements can be consistently merged with real elements (and without defining 2-types twice).

- We can then assert requirements on the 'real' and 'ghost' elements separately, using ordinary $\mathcal{C}^2$ -formulas.



- We have to take elaborate precautions to ensure that ghost elements can be consistently merged with real elements (and without defining 2-types twice).
- The exact complexity seems still to be open.

- In the previous lecture, we met the **fluted fragment**, $\mathcal{FL}$:

No lecturer introduces any professor to every student

$$\forall x_1 (\text{lectr}(x_1) \rightarrow \neg \exists x_2 (\text{prof}(x_2) \wedge \forall x_3 (\text{stdt}(x_3) \rightarrow \text{intro}(x_1, x_2, x_3))))).$$

- Remember, argument-sequences in predicates are strictly increasing, ending with the ‘most recently quantified’ variable.

- The fluted fragment does not allow us to say that a binary relation is reflexive or symmetric:

$$\begin{aligned} &\forall x_1 \, r(x_1, x_1) \\ &\forall x_1 \forall x_2 \, (r(x_1, x_2) \rightarrow r(x_2, x_1)) \end{aligned}$$

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- Of course, we can add reflexive or symmetric relations as we did with transitive relations.
- But maybe we can do better ...

- In the **adjacent fragment**, $\mathcal{AF}$, we consider index-normal sentences, and require that, in any atom, **adjacent arguments have indices differing by at most 1**. Equality is allowed.
- Thus,

$$\begin{aligned} &\forall x_1 r(x_1, x_1) \\ &\forall x_1 \forall x_2 (r(x_1, x_2) \rightarrow r(x_2, x_1)) \end{aligned}$$

are in the adjacent fragment, but

$$\forall x_1 \forall x_2 (r(x_1, x_2) \rightarrow \forall x_3 (r(x_2, x_3) \rightarrow r(x_1, x_3)))$$

is not.

- The following sentence is in $\mathcal{AF}$:

$$\begin{aligned} &\forall x_1 \forall x_2 \forall x_3 \exists x_4 \forall x_5 (p(x_1, x_2, x_3, x_2, x_3, x_4, x_5) \rightarrow \\ &\quad p(x_1, x_2, x_3, x_4, x_3, x_4, x_5)). \end{aligned}$$

- Denote by $\mathcal{AF}^m := \mathcal{AF} \cap \mathcal{FO}^m$ the *m*-variable adjacent fragment.
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 - Given a formula $\varphi \in \mathcal{AF}^m$ ($m \geq 3$), compute another formula $\psi \in \mathcal{AF}^{m-1}$ (involving new predicates), (finitely) satisfiable iff φ is.

- Denote by $\mathcal{AF}^m := \mathcal{AF} \cap \mathcal{FO}^m$ the *m-variable adjacent fragment*.
- We then prove a 'small' model property, following the same strategy as for the fluted fragment:
 - Given a formula $\varphi \in \mathcal{AF}^m$ ($m \geq 3$), compute another formula $\psi \in \mathcal{AF}^{m-1}$ (involving new predicates), (finitely) satisfiable iff φ is.
- This yields much the same complexity bounds as for $\mathcal{FL}$.

- Thus we obtain:

Theorem (Bednarczyk, Kojelis, P-H, 24)

$\mathcal{AF}$ has the finite model property; moreover, $\text{Sat}(\mathcal{AF}^{[m]})$ is in $(m-1)\text{-NEXPTIME}$, for $m \geq 2$.

- Without equality, this reduces to $(m-2)\text{-NEXPTIME}$, for $m \geq 3$.

- Can we go any further?
- No. We are at the end of the road.

Theorem (Bednarczyk, Kojelis, P-H, 25)

Adding any other allowed variable sequence to the allowed variable sequence of $\mathcal{AF}$ yields a fragment $\mathcal{L}$ such that $\text{Sat}(\mathcal{L})$ and $\text{FinSat}(\mathcal{L})$ are undecidable.

- But adding counting quantifiers to the adjacent fragment results in undecidability.

Theorem (Kojelis, 25)

$\text{Sat}(\mathcal{AFC}^4)$ is undecidable;
 $\text{Sat}(\mathcal{AFC}^3)$ is decidable.
 $\text{FinSat}(\mathcal{AFC}^3)$ is undecidable.



- The decidability proof for $\mathcal{AFC}^3$ yields no complexity bound.

- This course concerned the *Entscheidungsproblem*, as formulated by D. Hilbert and W. Ackermann.
- As explained, H. and A.'s question was answered negatively by A. Church and A. Turing.
- This turned the decision problem into a *classification problem*.
- We considered various fragments of first-order logic:
 - the monadic fragment,
 - the 2-variable fragment,
 - the guarded fragment,
 - the Gödel fragment,
 - the (one- and) two-variable fragment with counting,
 - the fluted fragment.
- We established the FMP (or its failure) and obtained complexity bounds for the (finite) satisfiability problems.
- If you want to know more, google us.